

Water Segmentation Using Multispectral Satellite Images

Technical Report

Introduction

Semantic segmentation is a fundamental task in computer vision that aims to classify every pixel in an image. In remote sensing, semantic segmentation is widely used for applications such as land cover classification, urban planning, flood monitoring, and water resource management

The objective of this project is to accurately segment water bodies from multispectral satellite imagery using deep learning models. Instead of relying on a single architecture, this project focuses on comparing multiple pretrained segmentation models and selecting the best-performing one based on quantitative evaluation metrics

Dataset

The project uses multispectral satellite images together with their corresponding binary segmentation masks

Before training, several preprocessing steps were applied

Reading multispectral GeoTIFF images

Computing global dataset statistics

Image normalization

Data augmentation using Albumentations

Training and validation split

Preparing PyTorch datasets and dataloaders

These preprocessing steps improve model generalization and stabilize the training process

Methodology

Unlike the previous week's implementation, this project evaluates multiple deep learning segmentation architectures

The following pretrained models were implemented and compared

U-Net with a pretrained ResNet34 encoder

DeepLabV3+ with a pretrained ResNet50 encoder

Satellite-pretrained U-Net using TorchGeo pretrained weights

Each model was trained using the same preprocessing pipeline, optimizer, loss function, and evaluation metrics to ensure a fair comparison

Initially, the encoder weights were frozen while only the decoder layers were trained. After the initial training stage, the encoder was unfrozen and the selected model was fine-tuned to improve segmentation performance

The objective was not only to train several models but also to identify the architecture that achieved the highest validation performance

Model Comparison

A systematic comparison was performed between all candidate models

Each architecture was evaluated using identical validation data and the following metrics

Dice Score

(Intersection over Union (IoU

Precision

Recall

F1 Score

Training and validation losses were monitored during training to evaluate convergence and detect overfitting

After completing the comparison, the best-performing model was selected according to the evaluation metrics. This model was then trained for additional epochs through fine-tuning to maximize segmentation accuracy

Results

The comparison demonstrated that pretrained models significantly outperform training from scratch

Among all evaluated architectures, one model achieved the highest Dice Score and IoU on the validation dataset. This model was selected as the final solution and further fine-tuned to improve overall segmentation quality

The final model produced more accurate water boundaries, faster convergence, and better generalization compared with the other evaluated models

Conclusion

This project demonstrates the effectiveness of transfer learning for multispectral semantic segmentation

Rather than relying on a single architecture, multiple pretrained segmentation models were compared under identical experimental settings. Selecting the best-performing model followed by fine-tuning resulted in improved segmentation accuracy and more reliable predictions

Future improvements may include experimenting with transformer-based segmentation networks, larger datasets, and advanced augmentation techniques